

Reading residuals

a4a Training Course — Module 2.2

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Why this module exists



Residuals are your main tool to manage your fit

If you cannot read them, you cannot configure the model.

We stay on the mean model from Module 2.1 — a fit we already know is bad — and learn what each tool can show us.

Three kinds of residual

Type	Divided by
deviances	nothing — the raw log difference
standardized	the residual standard deviation
pearson	the model's predicted square root variance

With `vmodel = ~1` the variance is one constant, so pearson and standardized look alike.

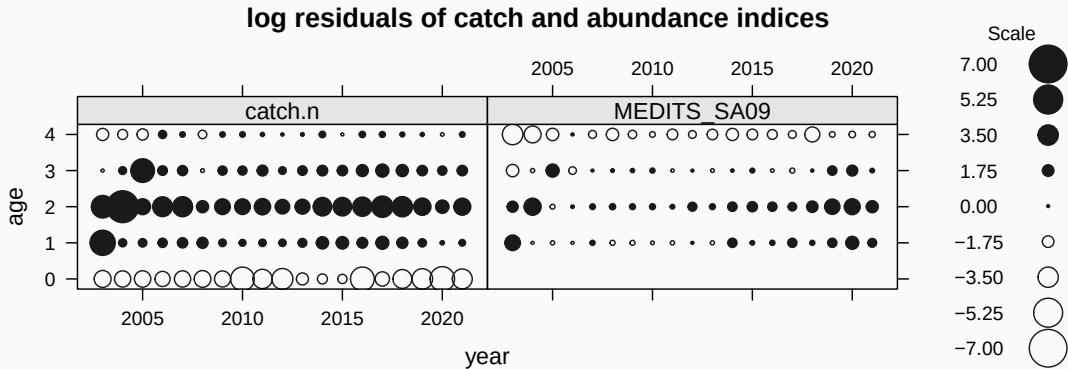
Each tool answers a different question

Tool	Question
<code>plot(res)</code>	does the fit drift through time ?
<code>plot(res, by="age")</code>	is it off at particular ages ?
<code>bubbles(res)</code>	same but with bubbles :-)
<code>qqmath(res)</code>	is the error distribution right?
<code>plot(fit, stk) & plot(fit, idx)</code>	check differences in patterns
<code>fitSumm(fit)</code>	number of parameters VS observations

Look by year **and** by age, every time. A pattern invisible in one view is often obvious in the other.

Rows, columns, diagonals

bubbles(res)

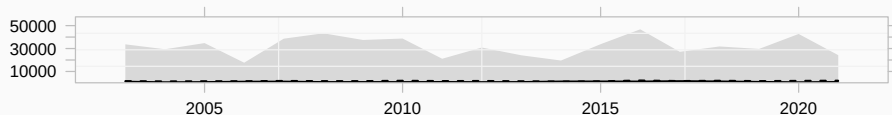


Does it reproduce the catch?

Everything so far was **at age**. This asks the blunter question, on aggregated catch in weight.

```
cd <- computeCatchDiagnostics(fit01, stk)
plot(cd, type = "prediction", probs = c(0.025, 0.975))
```

Prediction error



(shaded area = CI95%, dashed line = median, solid line = observed)

The observed catch sits well below the predicted interval — the mean model does not reproduce it at all.

Two cautions before you start

 Check these every time

convergence must be **0** in `fitSumm()`. Anything else and the rest of the output is meaningless.

AIC and BIC only compare models fitted to the **same data** — never across stocks or across different year/age ranges.

Now go and look at a bad model properly.

Exercise

Open `handson_M2.2_residuals.R`.

1. Compute all three residual types and compare their spread
2. Work through `plot`, `bubbles`, `qqmath`, `plot(fit, stk)`
3. Build the catch diagnostics
4. Decide, from the evidence, what you would change first

Then try it: refit with an age effect on F and see whether the thing you expected to improve actually did.